

George: Cold-Chain and Last-Mile Control Standard

Scope: Temperature-sensitive peptide-related products supplied in South Africa
Prepared by: Manus AI
Date: 18 August 2026

Professional disclaimer. This standard is an operational control framework, not a legal opinion, SAHPRA approval, product-registration decision, customs ruling, or certification that a peptide or route to market is lawful. The exact product, form, label, intended use, claims, schedule, customer and activity determine the applicable South African requirements. A “research use only” label does not by itself remove medicines, scheduled-substance, customs, transport, or other controls.

1. Control objective

George must preserve each product within its **manufacturer-approved and stability-supported conditions** from supplier release until receipt by an authorised consignee. There is no universal peptide storage temperature. Some products may be qualified for controlled ambient transport, some for refrigeration, and others for frozen or ultra-low conditions. The approved product master must therefore state the exact storage range, permitted transit range, freeze sensitivity, light and moisture protection, stability-supported excursion budget, qualified pack-out, logger requirements, and customer handling instructions.

For products governed as medicines, SAHPRA’s Good Wholesaling Practice guideline expects validation of ambient and cold-chain warehouse conditions, lagged containers, cold-chain processes, computerised systems, and transport systems. It also requires calibrated monitoring equipment, retained temperature records, controlled returns, and Responsible Pharmacist oversight of relevant disposition decisions.[1](#)

2. Non-negotiable release gates

A temperature-sensitive shipment may not be booked until the following controls are approved.

Gate	Required evidence	Release authority
Product and legal route	Approved SKU classification, product authority, activity/licence/permit assessment, authorised customer type and approved claims	Regulatory/Legal and Responsible Pharmacist where applicable

Gate	Required evidence	Release authority
Temperature specification	Labelled storage range, permitted transport range, freeze/heat/light/moisture sensitivity and stability-supported excursion information	Quality
Packaging	Approved bill of materials, conditioning instructions, pack-out diagram, payload range, logger position, tamper closure and qualified duration	Quality
Lane	Approved origin, gateway, ground handler, customs broker, storage points, domestic carrier, last-mile route, receiving site and contingency	Quality and Logistics
Service providers	Qualified manufacturer, forwarder, airline/handler, importer, warehouse, courier and subcontractors under quality agreements	Quality and Procurement
Customs	Importer, HS and valuation review, permit/licence pack, invoice, packing list, transport document, origin evidence and inspection/recovery instructions	Customs owner and broker
Recipient	Authorised consignee, verified delivery address, delivery window, trained receiving contact and available compliant storage	Customer Operations
Shipment release	Correct product, lot, expiry, quantity, remaining shelf life, condition, seals, logger, labels and complete documents	Authorised Quality/Responsible Pharmacist

3. Service-class requirements

Service class	Typical use	Minimum control set
Qualified ambient	Product with approved ambient transit and storage range	Qualified lane or vehicle; protection from heat, sunlight, moisture, contamination and theft; documented maximum dwell; tamper-evident closure; monitoring based on risk
Refrigerated	Exact approved range, commonly but not universally 2–8°C	Qualified passive or active system; conditioned phase-change material; freeze barrier; calibrated logger; seasonal lane evidence; immediate controlled receipt
Frozen or ultra-low	Exact approved range	Qualified active or dry-ice/passive system; sublimation or energy reserve; redundant monitoring for high-risk lanes; approved replenishment/recovery plan; dangerous-goods compliance where triggered
Secure overlay	Scheduled, high-value, diversion-prone or sensitive stock	Restricted information, vetted personnel, locked storage/vehicle, GPS/geofencing, serialised seals, no unattended delivery and defined loss escalation

Service class	Typical use	Minimum control set
Dangerous-goods overlay	Dry ice, certain lithium-powered trackers, solvents or other classified components	Current air/road/sea classification, trained staff, approved packaging, marks, labels, documentation, segregation and carrier acceptance

The strictest applicable controls are cumulative. A refrigerated product containing dry ice or a lithium-powered tracker may require refrigerated, dangerous-goods, and secure overlays simultaneously.

4. Packaging qualification

WHO technical guidance requires insulated passive containers to be qualified as complete systems, including temperature-stabilising media, partitions, dunnage, payload, thermal mass, conditioning, monitor position, route profile, and duration through arrival in the recipient's controlled store.² George's qualification protocol should include the following elements.

Qualification element	Acceptance requirement
Product profile	Exact approved range and stability-supported limits; freeze sensitivity explicitly identified
Bill of materials	Shipper, liners, phase-change media/gel packs/dry ice, spacers, absorbent material, dunnage, seals and labels uniquely identified
Conditioning	Written time and temperature instructions for every coolant component; trained packer verification
Payload	Minimum and maximum quantity, mass and configuration challenged; void-fill method specified
Thermal profiles	Representative or worst-case South African summer and winter ambient profiles, source-country conditions and unconditioned handoffs
Duration	Planned door-to-door time plus a documented delay margin covering at least the credible missed connection, customs hold or failed-delivery scenario selected by risk assessment
Freeze protection	Product never directly contacts frozen media unless qualification expressly supports it; freeze alarm and barrier placement challenged
Monitor position	Hot/cold or otherwise most representative position established by qualification; redundant device used where risk warrants it
Physical stress	Closure, tamper evidence, vibration, drop, leakage and pressure risks addressed for the mode and product

Qualification element	Acceptance requirement
Records	Approved protocol, raw data, deviations, report, authorised approval, version-controlled pack-out and requalification triggers

A pack-out is not interchangeable across payloads or routes unless qualification supports the range. Any change to shipper, coolant, supplier, packaging component, payload range, logger position, carrier, route, schedule, hub or product condition requires documented change assessment and possible requalification.

5. Lane qualification

The qualified lane begins at the supplier's controlled dispatch point and ends only when the product is placed into the recipient's approved storage. WHO advises that route qualification use weather data, laboratory testing and field trials appropriate to the anticipated conditions.²

The lane profile must record origin pickup, export staging, airport or port dwell, airline/ground-handler storage, customs and other-government-agency inspection, controlled transfer, receiving warehouse, domestic line haul, courier hub, delivery route, delivery window, and recipient put-away. Qualification should challenge the highest-risk seasonal profile, minimum and maximum payload, weekend/public-holiday effects, customs examination, missed connection, hub delay, vehicle breakdown, load-shedding or power interruption, absent recipient and return-to-depot route.

The approved lane file should state maximum dwell at each node, permitted storage condition, required pre-alert, handoff evidence, emergency contacts, recovery facility, coolant replenishment rule, decision point for return or destruction, and insurance/security responsibilities.

6. Temperature monitoring and calibration

SAHPRA GWP requires calibrated measuring equipment under an approved schedule and temperature records available for review. It recommends monitoring positions based on temperature mapping and retention of monitoring records for at least product shelf life plus one year, or longer where national legislation requires.¹ WHO recommends calibration against a certified traceable reference at least annually unless otherwise justified, while single-use devices supplied with a manufacturer's calibration certificate need not be recalibrated.²

Every temperature-controlled consignment must therefore have:

- 1 A logger uniquely linked to the shipment, product lots, pack-out and custody record.
- 2 A valid calibration certificate or manufacturer certificate suitable for the device type.

- 3 The correct range, sample interval, start delay, time zone and alarm limits configured under an approved procedure.
- 4 Placement exactly as established by packaging qualification.
- 5 Raw electronic data retained with device ID, calibration status, start/stop times, audit trail and reviewer.
- 6 A documented rule that **any alarm, missing file, stopped logger, time gap, implausible trace or custody gap triggers QUALITY-HOLD.**
- 7 Redundant monitoring or real-time tracking for frozen, high-value, long-duration or difficult-to-recover shipments where risk assessment requires it.

Active or transmitting devices used in air cargo require airline acceptance. Lithium-battery trackers must meet the applicable current IATA Dangerous Goods Regulations, operator variations and technical requirements.[3](#)

7. South African import and customs handoffs

SARS requires imported goods to be declared and may examine goods, request samples, require permits, or detain them for other government departments.[4](#) Cold-chain design must therefore assume that inspection can occur and must prevent an uncontrolled opening or thermal failure.

Before origin dispatch, the appointed broker must confirm importer registration, product description, HS treatment, valuation and origin basis, required SAHPRA or other authorisations, commercial invoice, packing list, airway bill/bill of lading, permits, gateway and inspection contact. The shipment instruction must identify the external transport range, 24-hour contact, controlled-storage requirement, latest safe release time, seal and logger handling, authorised opening/repacking process, and recovery provider.

Where customs or another authority opens the pack, the handoff record must capture time, officer or handler, seal broken, logger status, product exposure, replacement pack-out, new seal, storage location and onward release. Spare qualified packaging, coolant and loggers should be available at the gateway or through a prearranged recovery provider.

For air cargo booked as time- and temperature-sensitive healthcare cargo, IATA requires the Time and Temperature Sensitive label to display the external transportation range and uses an acceptance checklist for airlines and ground handlers.[3](#) The airway bill, package label and booking must show consistent temperature information.

8. Chain of custody

The shipment record must support one-up/one-down lot traceability and unambiguous custody from supplier to recipient.

Required field	Control purpose
Shipment and package IDs	Link every handling unit to the order and documents
SKU, lot, expiry and quantity	Enable release, reconciliation and recall
Seal and logger IDs	Detect tampering and link environmental evidence
Custodian, organisation and named handler	Establish accountability
Place, timestamp and vehicle/flight	Reconstruct route and dwell
Storage condition and package condition	Demonstrate preservation and identify deviations
Reason for handoff or delay	Explain non-standard movement
Recipient name, role and signature/OTP	Prove delivery to an authorised person

Scans or signed records are required at origin pickup, export acceptance, airline/ground-handler handoff, customs release, controlled transfer, warehouse receipt, domestic dispatch, courier hub where applicable, last-mile departure and final delivery. Unapproved subcontracting is prohibited; every custody party must be included in the qualified service chain or pre-approved under change control.

9. Last-mile delivery standard

The last mile is a controlled extension of the warehouse, not an ordinary parcel service. The approved courier must operate only on qualified routes, vehicles or validated parcel shippers and must meet written quality and service-level requirements.

9.1 Pre-dispatch controls

The warehouse must verify that the lot is RELEASED, not expired or recalled, meets the recipient's minimum remaining shelf life, matches the authorised order and customer class, and uses the approved pack-out. The delivery address and recipient may not be changed informally. Dispatch requires a pre-alert with estimated arrival, storage range, urgency of put-away, package/temperature-indicator instructions and escalation contact.

9.2 Courier and vehicle controls

The courier agreement must identify approved subcontractors, hubs, vehicle standards, cleaning, pest/contamination prevention, lock and security, route monitoring, temperature control where the vehicle is the qualified environment, logger handling, maximum dwell, priority recovery, deviation notification, records, training, audits, CAPA, recall assistance and business continuity. High-value or scheduled products require vetted drivers, GPS/geofencing, tamper-evident seals, restricted route information and no unattended stops or safe drops.

9.3 Delivery acceptance

Delivery must be made only to the authorised address and a trained recipient with immediate access to approved storage. Proof of delivery should capture arrival time and geolocation where lawful, recipient identity and role, signature or secure OTP, package and seal condition, quantity, logger/indicator handling, receiving temperature where the SOP requires it, and time placed into controlled storage.

The courier may not leave a temperature-sensitive shipment at reception, a security gate, outside a building, in an unattended locker or at an alternative address unless that exact handoff was qualified and approved. No “safe drop” is permitted.

9.4 Failed delivery

A failed delivery triggers an immediate exception call and a predefined decision tree. The courier must not retain the shipment in an uncontrolled vehicle or hub. It returns directly to an approved qualified depot or warehouse while preserving the validated condition and custody record. The lot is placed into RETURN-QUARANTINE or QUALITY-HOLD pending authorised assessment. Redelivery may occur only when the remaining qualified duration, logger data, custody, package integrity and recipient availability support it.

10. Excursion and deviation management

An alarm is not automatically a rejection, and a later normal reading is not a release. The following sequence is mandatory:

- 8 Identify the affected shipment, package, SKU, lots, quantity and exposure interval.
- 9 Place affected stock on electronic and physical QUALITY-HOLD.
- 10 Preserve it under the approved storage condition without discarding logger or packaging evidence.
- 11 Secure raw logger data, custody records, ambient data, packaging version, calibration evidence, route timeline and handler statements.
- 12 Assess magnitude, duration, freeze/heat/light exposure, cumulative prior excursions and product stability information.
- 13 Consult the manufacturer, marketing-authorisation holder or appropriately qualified product expert where required.

- 14 Record authorised disposition: release, reduced shelf life where legally and scientifically supported, return, reject or destroy.
- 15 Investigate root cause and approve CAPA before the same configuration is used again where the deviation indicates loss of control.

Customer service, Sales, Procurement, the courier and the warehouse operator may not independently release an excursion-affected lot.

11. Returns, recalls and destruction

SAHPRA GWP requires returned and rejected products to be clearly labelled, sealed and stored separately in a secure designated area. The Responsible Pharmacist is responsible for the final decision to return relevant goods to stock or destroy them.¹ For temperature-sensitive returns, George should apply a default **no-restock** policy unless complete evidence demonstrates original unopened and intact packaging, correct identity, acceptable shelf life, continuous custody, no adverse exposure and an authorised quality assessment.

Recalled stock must remain under the required storage condition during storage and transport until final disposition. Records must permit customer and quantity reconciliation, and the final recall report must reconcile delivered and recovered quantities.¹ Destruction must use an authorised waste route appropriate to the product and schedule, with sealed transfer, manifest, witnessing where legally required, quantity reconciliation and certificate retained in the quality and stock systems.

12. Facility, equipment and power controls

Temperature-controlled storage must be mapped under representative operating conditions and seasonally where appropriate. Monitoring probes must be placed at identified hot and cold points, not merely near the door or control display. Refrigerators, freezers and cold rooms require continuous monitoring where product risk warrants it, alarms with 24-hour escalation, calibrated devices, preventative maintenance, restricted access, capacity management and contingency transfer space.

Backup power must be sized and tested for the critical load and credible outage duration. The continuity plan should state UPS/generator changeover, fuel, alarm continuity, manual monitoring, maximum safe downtime, emergency transfer vehicle, alternate store, authorised decision-maker and priority product list. Load-shedding must be treated as a planned operational condition in South African risk assessment, not an unforeseen emergency.

13. Required records

George must retain the approved product temperature profile, stability/excursion basis, pack-out qualification, lane qualification, service-provider approval, calibration, training, shipment booking, customs pack, chain of custody, logger raw data, receipt review, release or hold decision, deviation/CAPA, failed-delivery record, return disposition, recall reconciliation and

destruction evidence. For medicines under the SAHPRA GWP framework, temperature-monitoring data should be retained for at least the product shelf life plus one year or longer if required by national legislation.[1](#)

14. Minimum courier SLA

SLA metric	Initial control target
Approved lane, pack-out and recipient before dispatch	100%
Correct delivery address and authorised recipient	100%
Temperature-compliant delivery with readable complete evidence	≥99.5%
Complete required custody scans and seal reconciliation	100%
Unauthorised safe drops	0
Pre-clearance completed before international dispatch	100%
Critical/frozen alarm or loss notification	Within 1 hour of awareness; tighter where risk requires
Failed-delivery recovery to qualified location	Within remaining qualified duration and route SLA
Returns placed into quarantine on receipt	100%
Quality disposition before redelivery or restock	100%
Theft, tampering or unexplained loss	0; immediate escalation and investigation
Records supplied for complaint or recall	Within the contractual rapid-response window

Targets are internal control standards and must be adjusted to product risk, licence conditions and contractual feasibility. A target does not permit silent acceptance of the remaining percentage; every exception requires investigation and disposition.

15. Pre-dispatch checklist

Check	Pass requirement
SKU and legal route	Approved and in scope
Lot status	RELEASED; not held, recalled, rejected, expired or short-dated
Customer and address	Authorised and verified
Pack-out	Correct approved version, payload and conditioning
Logger	Correct device, calibration, settings and placement
Seals and labels	Recorded; temperature range consistent across booking, airway bill and label
Documents	Invoice, packing list, permits/authorities, transport and emergency instructions complete
Lane and capacity	Booking, controlled hubs, receiving capacity and contingency confirmed
Recipient	Pre-alert acknowledged; storage ready
24-hour contacts	Supplier, forwarder, broker, handler, warehouse, courier and Quality reachable

16. Receipt checklist

Check	Required action
Arrival and custody	Record actual time, courier, recipient and seal status
Package	Inspect damage, wetness, contamination, tampering and coolant condition
Product	Verify SKU, lot, expiry, quantity and label
Temperature evidence	Stop/download/read logger according to SOP; retain raw data

Check	Required action
Alarm or evidence gap	Immediate QUALITY-HOLD; no verbal release
Put-away	Move promptly to approved storage; record exact bin and time
Documents	Reconcile shipment, customs, CoA and delivery evidence
Release	Only authorised Quality/Responsible Pharmacist disposition changes status

17. Sources